1.3.23:

Solution:

(a) *Proof.* Let W_1, W_2 be subspaces of a vector space V. We must first show that $W_1 + W_2$ is a subspace. This is obvious as $0 \in W_1$ and $0 \in W_2$ so $0 \in W_1 + W_2$ by definition. As well, for $x = x_1 + y_1$ and $y = x_2 + y_2$ in $W_1 + W_2$ (with $x_1, x_2 \in W_1$ and $y_1, y_2 \in W_2$) and $c \in \mathbb{F}$ we have $cx = cx_1 + cy_1 \in W_1 + W_2$ and $x + y = (x_1 + x_2) + (y_1 + y_2) \in W_1 + W_2$ because W_1 and W_2 are closed under vector addition and scalar multiplication. Hence, by Thm. 1.3 $W_1 + W_2$ is a subspace. The fact that $W_1 \subseteq W_1 + W_2$ and $W_2 \subseteq W_1 + W_2$ follows clearly from the fact that $0 \in W_1 + W_2$ so for $x \in W_1, x = x + 0 \in W_1 + W_2$ and for $y \in W_2, y = 0 + y \in W_1 + W_2$. □

(b) *Proof.* Consider vector space V with subspaces W_1, W_2, W_3 such that $W_1 \subseteq W_3$ and $W_2 \subseteq W_3$. It follows that $W_1 + W_2 \subseteq W_3$ since for all $x, y \in W_3$ we know that $x + y \in W_3$ as well, so by restricting $x \in W_1$ and $y \in W_2$ it is clear that $W_1 + W_2 \subseteq W_3$. □

1.3.24:

Solution:

Proof. Clearly $W_1 \cap W_2 = \left\{\vec{0}\right\}$ since if $x = (x_1, \ldots, x_n) \in W_1 \cap W_2$ then $x_n = 0$ (as $x \in W_1$) and $x_1 = \cdots = x_{n-1} = 0$ (as $x \in W_2$), so $x = \vec{0}$. It is also clear that $W_1 + W_2 = \mathbb{F}^n$ since any $x = (c_1, c_2, \ldots, c_n) \in \mathbb{F}^n$ can be represented by

$$(c_1, c_2, \ldots, c_{n-1}, 0) + (0, \ldots, 0, c_n) \in W_1 + W_2$$

so $\mathbb{F}^n \subseteq W_1 + W_2$ and $W_1 + W_2 \subseteq \mathbb{F}^n$ by definition of subspaces. Thus $W_1 \oplus W_2 = \mathbb{F}^n$. □

1.3.25:

Solution:

Proof. It is clear that $W_1 \cap W_2 = \left\{\vec{0}\right\}$ since if $f(x) = a_n x^n + \cdots + a_1 x + a_0 \in W_1 \cap W_2$ then $a_i = 0$ for all even i (as $f \in W_1$) and $a_i = 0$ for all odd i (as $f \in W_2$), so every coefficient of f is 0 and f is the zero polynomial. As well, clearly $W_1 + W_2 = P(\mathbb{F})$ as any $h(x) = c_k x^k + c_{k-1} x^{k-1} + \cdots + c_1 x + c_0$ can be represented as a sum of some $f(x) \in W_1$ and $g(x) \in W_2$ where f is the polynomial formed by summing all the terms of odd degree of h and g is the polynomial formed by summing all the terms of even degree of h. So, $P(\mathbb{F}) \subseteq W_1 + W_2$. Thus $W_1 \oplus W_2 = P(\mathbb{F})$ (since $W_1 + W_2 \subseteq P(\mathbb{F})$ by definition). □

1.3.30:

Solution:

Proof. ($\Leftarrow$) Assume that each vector in V can be uniquely written as $x_1 + x_2$ with $x_1 \in W_1$ and $x_2 \in W_2$. Since this representation is unique and $0 \in W_1$ and $0 \in W_2$ it follows that $W_1 \cap W_2 = \left\{\vec{0}\right\}$ since for any $z \in W_1 \cap W_2$ we have $z = z + 0 = 0 + z$ (once with $z \in W_1, 0 \in W_2$ and once with $0 \in W_1, z \in W_2$), so uniqueness forces $z = 0$. It follows from our assumptions that $W_1 + W_2 = V$ since every vector x has a unique representation of the specified form. So, $W_1 \oplus W_2 = V$
($\Rightarrow$) Conversely, assume that V is the direct sum of W_1 and W_2. Existence is clear as $W_1 + W_2 = V$ by definition. Uniqueness follows as if $x_1 + x_2 = y_1 + y_2$ then $x_1 - y_1 = y_2 - x_2 \in W_1 \cap W_2 = \left\{\vec{0}\right\}$ forcing $x_1 = y_1$ and $x_2 = y_2$. □